Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Building compact wearable interfaces for dexterous teleoperation			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		*Information & Communication*	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	Robot Control and Intelligence Technologies	Digital Content	Virtual Reality
Concept of Proposed Technology	The proposed technology is a compact, wearable teleoperation interface designed to support physical AI systems that require precise, contact-rich interaction with the physical world. The system combines high-density fingertip haptic feedback, hand posture sensing, and real-time tactile rendering to translate robot contact signals into perceptually meaningful cues for a human operator. At the hardware level, the platform uses arrays of miniaturized vibrotactile actuators (9–12 LRAs per fingertip) embedded in soft silicone thimbles that are lightweight, low cost, and suitable for repeated use. The core technical contribution lies in advanced driving and rendering algorithms that operate at both the individual actuator level and across actuator arrays. Temporally shaped pulse inputs and coordinated inter-actuator timing are used to manage vibration propagation on the skin, enabling localized rendering of contact events such as contact onset and location, force change, shear direction, and slip tendency. These tactile cues are synchronized with hand tracking (≥60 Hz) and robot state for human-in-the-loop teleoperation. By improving operator situational awareness during dexterous manipulation, the system enables more accurate teleoperation and higher-quality human demonstration data, supporting robot learning,			

	manipulation research, and physical AI development.
Rationale for Joint R&D Project	Advancing physical AI systems for dexterous manipulation requires both high-quality human interaction data and reliable interfaces that can operate in real-world settings. While academic research has demonstrated high-resolution tactile feedback using novel actuation approaches, many such systems face challenges in durability, scalability, and integration. Conversely, existing commercial wearable systems are robust and manufacturable but lack expressive haptic capabilities and rendering strategies needed for contact-rich teleoperation. This joint R&D project addresses this gap through a complementary partnership. The research institute contributes expertise in haptic perception, actuator-level and array-level driving algorithms, and system-level evaluation of teleoperation performance. These contributions focus on how robot contact information should be encoded and rendered on the human hand to improve manipulation accuracy and data quality. Korean SME partners contribute strengths in wearable system engineering, hand tracking integration, wireless playback, control electronics, and reliability-driven product development. Collaboration is essential to co-design rendering algorithms and hardware architectures that scale together, ensuring that perceptual performance, robustness, and manufacturability are addressed simultaneously. This integrated approach reduces translational risk and enables the development of a near-commercial wearable teleoperation platform that directly supports Physical AI training, evaluation, and deployment.
Keywords	1. Wearables 2. Dexterous teleoperation 3. Dexterous manipulation 4. Haptics 5. Tactile Feedback

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	**Start of the Project**	Level (2)	**End of the Project**	Level (5)
R&D Objectives	Although recent humanoid robots can perform large-scale motions such as walking and jumping, they still struggle with fine, contact-rich dexterous manipulation. Recent studies have shown that dexterous teleoperation can efficiently generate high-quality manipulation data by leveraging human skill (1-2). However, most existing teleoperation systems rely primarily on visual feedback, which is often insufficient for complex manipulation. For contact-rich, complex dexterous manipulation tasks, operators need high-fidelity **tactile feedback** that conveys precise contact information such as location, orientation, curvature and texture. Several high-density wearable tactile gloves have been proposed to deliver rich contact information using novel actuation methods, such as microfluidic channels (6-7) or electroosmotic pump (5) (Fig. 1). However, their limitations in long-term reliability and stability make them difficult to deploy in practical, product-level systems.			

Figure 1. Commercialized high-density tactile gloves

The objective of this R&D effort is to develop a **compact and reliable wearable interface for dexterous teleoperation** that 1) delivers high-fidelity tactile feedback to the operator and 2) captures precise hand posture and interaction states, enabling robust data collection and improved teleoperation performance. Our longer-term goal is to mature the system toward a **deployable, production-level platform** through close interdisciplinary collaboration in hardware, sensing, and learning.

As a foundation for this project, our lab at UIUC has developed **a compact and reliable wearable fingertip haptic platform** that provides localized tactile feedback. The device integrates an array of miniaturized vibrotactile actuators (linear resonant actuators, LRAs) embedded in a soft silicone fingertip thimble (Fig. 2A). The primary technical contribution of this platform lies in the **driving and rendering algorithms** used to control these actuators. We design stimulation at both the individual actuator level, using temporally shaped drive pulses, and at the array level, by coordinating timing across actuators to manage vibration propagation on the skin. This approach enables improved spatial localization using brief, precisely timed pulses rather than continuous vibration (Fig. 2B). By pairing these algorithms with commercially mature actuators, the system remains robust, **easy to put on and take off**, low cost (approximately **$20 per fingertip module**), and preliminarily validated over more than **100,000 actuation cycles**. In collaboration with Korean small and medium-sized enterprises, this foundation will be scaled into a fully integrated wearable interface for dexterous teleoperation.

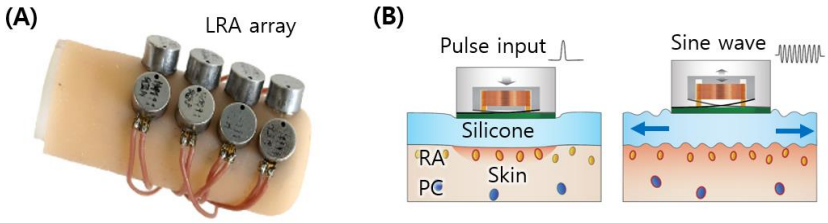

Figure 2. (A) Wearable fingertip haptic device using LARs, (B) Principle of providing localized feedback using pulse input

Specific R&D objectives

1. **Compact, high-resolution tactile feedback:** Increase the spatial resolution and perceptual clarity of fingertip feedback while maintaining a lightweight, comfortable, and wearable form factor with low-cost fabrication.
2. **Reliable sensing for teleoperation and data capture:** Integrate sensing to estimate **hand posture**, supporting consistent data collection across diverse manipulation tasks.
3. **System-level validation in dexterous teleoperation:** Demonstrate improved operator performance on contact-rich tasks (e.g., precise grasp adjustment, sliding/rolling, in-hand manipulation) and generate scalable datasets for robot learning.
4. **Toward a deployable platform:** Improve robustness, repeatability, and usability (comfort, setup time, maintainability) to move the system closer to a production-ready teleoperation tool.

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R&D Target Specifications

The following target performance specifications define the measurable outcomes to be achieved through this R&D project. Targets are organized by subsystem (tactile output, sensing/hand tracking, rendering/control, and deployability) and will be verified through bench testing, perception studies, and teleoperation evaluations (VR + real robot).

Wearable tactile feedback module

Item	Target Specification	Measurement/Verification
Actuator density	9–12 LRAs per fingertip (single fingertip module)	Hardware inspection BOM + layout check
Spatial pitch	Center-to-center spacing ≤ 4.5–6.0 mm	PCB/layout measurement
Durability	≥ **100,000 actuation cycles** per actuator without functional failure	Automated cycling test

Hand posture sensing

Item	Target Specification	Measurement/Verification
Hand tracking update rate	≥ **60 Hz** posture update	Sensor/SDK logging
Joint angle accuracy	Mean absolute error ≤ **10°**	Ground-truth comparison
Fingertip pose stability	Drift ≤ **2 mm** over 60 s static hold	Pose logging in controlled task

Rendering + control software

Item	Target Specification	Measurement/Verification
Rendering modes	At least **3 task-relevant cues**	Feature-to-pattern mapping checklist
Calibration time	User-specific calibration ≤ **3 minutes**	Timed setup procedure
Crosstalk control	absolute actuator identification > **80%**	Psychophysics study under standardized stimuli

Rendering + control software

Item	Target Specification	Measurement/Verification
Full system mass	≤ **350–450 g** (one-hand glove)	Weigh assembled system
Setup time	≤ **10 minutes** from unpack → ready-to-teleoperate	Timed protocol
Prototype cost target	full glove prototype BOM target ≤ **$300–$600**	BOM review

R&D Plan

To achieve our main goal of **building a compact and reliable wearable interface for dexterous teleoperation**, we have developed an R&D plan for both Phase 1 (Seed round) and Phase 2 (Full funding), as summarized in Figure 2.

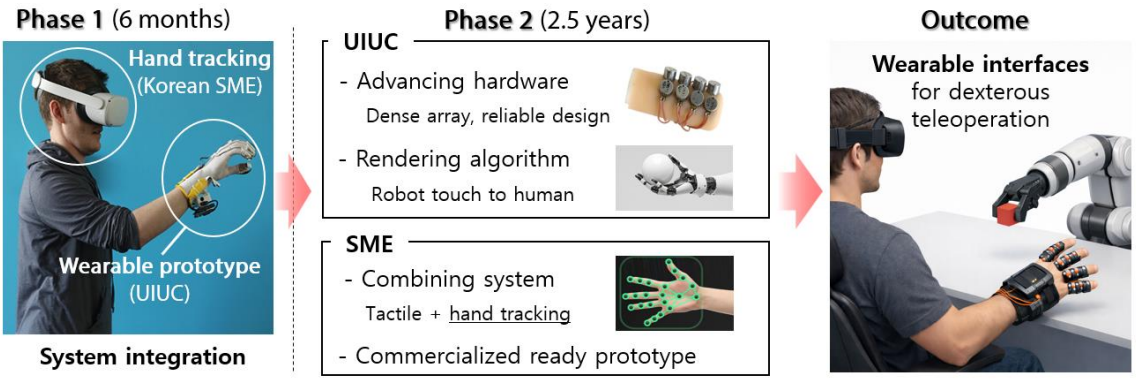

Figure 2. Overall R&D Plan during Phase 1 and Phase 2

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Phase 1 (6 months)

During Phase 1, our primary goal is to coordinate with a Korean SME partner to align our visions and define a detailed technical plan for Phase 2. We expect the SME partner to have strong experience in developing wearable interfaces, possibly starting from simpler haptic feedback systems, and in integrating hand tracking methods.

As stated earlier, we recently built a compact wearable tactile device with a 2 by 4 grid of small vibration motors (linear resonant actuators) (Fig. 1B). Each actuator is 6 mm in diameter and 4 mm tall. We found that simple continuous vibration does not produce well localized sensations because of it spreads on skin easily. However, we can provide more localized tactile feedback by driving the actuators with brief pulses.

Through this collaboration, we will provide a working prototype of our wearable tactile interface to the SME partner. The SME partner will then focus on integrating our system with their existing hand tracking and wireless playback platform and demonstrating a very simple robot teleoperation scenario to validate feasibility and highlight the potential impact of Phase 2.

Phase 2 (2.5 years)

During Phase 2, our goal is to build wearable interfaces for dexterous teleoperation that are close to commercialization level. To achieve this goal, we have defined clear milestones, as shown in Table 1. These milestones will be refined after further discussions with an SME partner.

Phase 2 - Stage 1 (1 year)

In Stage 1, we will focus on advancing the hardware, primarily by replacing the current actuators with smaller vibration motors and increasing the array density, for example, nine to twelve actuators per fingertip. We will implement a flexible PCB to simplify wiring and integration of high-density actuator arrays. To make the system easier to put on and take off and more robust, we will test multiple mounting methods.

In addition, we will conduct a perception study to improve localization feedback. Based on these results, we will optimize the hardware design parameters, including the material and thickness of the base soft silicone membrane and the input voltage profiles for each actuator with adaptive timing to reduce crosstalk. By the end of Stage 1, we expect to deliver an enhanced wearable tactile device with higher density and improved localization performance. In parallel, the SME partner will develop a full five finger prototype, including the controller and integration with their hand tracking and wireless playback system.

Table 1. R&D Plan during Phase 2

		Main task	Main outcomes
Stage 1 (1year)	UIUC	- Hardware upgrade (denser array, flexible PCB) - Perception study (localization)	- Wearable tactile device - Optimized localization methods
	SME	- Hand tracking and wireless playback integration - Muli-finger prototype and controller	- Tactile & hand-tracking prototype.
Stage 2 (1year)	UIUC	- Rendering algorithms (robot contact to operator) - Evaluation of the performance	- Optimized rendering algorithm
	SME	- Final product design and fabrication - Custom control board (10 fingertips)	- Commercialized ready prototype.
Stage 3 (6months)	UIUC	- System integration - Teleoperation evaluation	- Final system with validated teleoperation performance
	SME		

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Phase 2 - Stage 2 (1 year)

In Stage 2, we will develop **rendering algorithms** that translate robot contact signals into tactile patterns that are perceptually clear and operationally useful during dexterous teleoperation. Specifically, we will process robot contact data to extract task relevant features such as contact onset and location, force change, shear direction, and slip tendency, and then map these features to spatiotemporal actuator patterns on the wearable tactile device. The rendering will be designed to maximize localization while minimizing crosstalk on the skin by leveraging brief pulses and optimized timing, informed by our Stage 1 perception study. We will further develop task dependent rendering strategies that prioritize different cues depending on the manipulation context, and incorporate user specific calibration to improve interpretability and comfort. The resulting algorithms will be evaluated in representative teleoperation tasks by measuring objective manipulation performance and subjective user feedback.

In parallel, we expect the SME partner to focus on bringing the system closer to a commercialization ready product by improving reliability and long-term stability, developing a custom control circuit board to drive all ten fingertip wearable haptic modules (potentially 120 actuators), and integrating a complete hand tracking and wireless playback system. In addition, SME will also prepare the control firmware, so that they can easily implement the rendering algorithm developed from our lab at UIUC.

Phase 2 - Stage 3 (6 months)

In Stage 3, we will focus on integrating the rendering software developed at UIUC with the wearable hardware developed by our Korean SME partner. We will first validate the effectiveness of the complete wearable interface—combining tactile feedback, hand tracking, and rendering—through object manipulation tasks in a VR environment. We will then evaluate system performance during teleoperation of a physical robotic platform. Finally, we will benchmark our glove against higher cost commercial products (e.g. the Manus (8) glove or SenseGlove (9)), which provides hand tracking with simple vibrotactile or braking feedback, comparing performance across key metrics, such as task completion time, object drops, and cognitive load measures.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute

As outlined in the R&D plan, the primary role of the research institute at UIUC is to lead the research-intensive tasks—namely, developing the wearable hardware prototypes and the tactile rendering algorithms—and to transfer these outcomes to our Korean SME partner. Specifically, UIUC will (1) **design and optimize the wearable tactile interface** through iterative mechanical design and human perception studies, and (2) **develop and refine rendering algorithms** and software that translate robot contact information into perceptually meaningful tactile feedback for the operator.

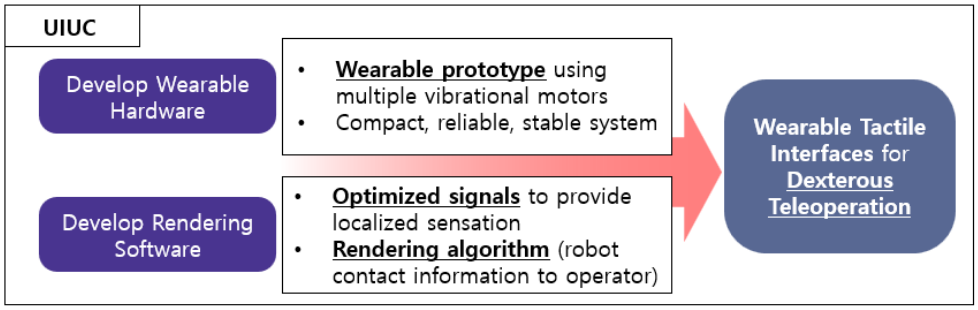

Figure 3. Expected R&R of UIUC

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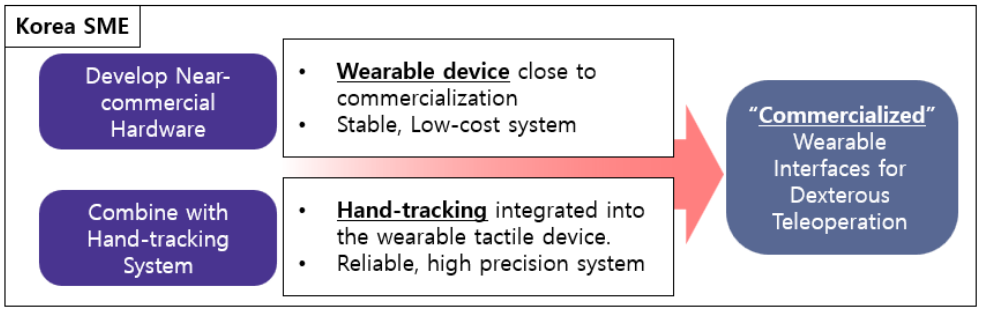
Expected R&R of Korean SME Partner	The primary role of the Korean SME partner will be to lead system integration and product-oriented development toward near-commercial hardware. The SME partner will receive the optimized wearable prototype design and the rendering algorithms from UIUC, and will focus on: (1) **developing a commercialization-ready wearable tactile interface** with reliable and stable actuation and low-cost fabrication, and (2) **building the fully integrated system** by combining the hand-tracking module, control hardware, and UIUC's rendering software to deliver an end-to-end wearable teleoperation interface.  Figure 4. Expected R&R of Korean SME Partner

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Outcomes** By the end of this project, we expect to deliver the following tangible outcomes: **1. Near-commercial Wearable teleoperation toolkit** • A compact, comfortable, and reliable wearable system that combines high-density fingertip tactile feedback with hand tracking, suitable for dexterous teleoperation. • A system integration package that includes hardware modules (multi-finger prototype), control electronics, and an operating software interface (API). **2. Validated tactile rendering software stack** • A set of real-time tactile rendering algorithms and reference software developed at University of Illinois Urbana-Champaign that translate robot contact signals (e.g., contact onset and location, force change, shear/slip cues) into perceptually clear spatiotemporal patterns. **3. Quantified performance gains in VR and real-robot teleoperation** • System-level validation showing measurable improvements in manipulation performance (e.g., higher success rate, fewer slip/drop events, faster correction of grasp errors) in both VR object manipulation and physical robot teleoperation. • Benchmark results against a commercial baseline glove such as Manus, demonstrating the advantage of high-fidelity tactile feedback beyond simple vibrotactile cues. **Expected Impact** **1. Scientific and technical impact** • Advances the state of wearable haptics for dexterous teleoperation by addressing two key barriers: localized tactile rendering (despite vibration propagation limits) and system-level reliability for repeated use. • Establishes a practical framework for mapping complex robot contact dynamics to interpretable human tactile cues—bridging hardware, perception, and real-time rendering. **2. Impact on teleoperation performance and operator capability**

	• Improves operator situational awareness during contact-rich manipulation (e.g., grasp adjustment, sliding/rolling, in-hand manipulation), enabling more precise control with reduced reliance on vision alone. • Reduces cognitive load and enhances confidence during fine manipulation through actionable tactile cues. **3. Economic and translational impact** • Strengthens collaboration between U.S. academic research and Korean SME engineering capabilities, accelerating translation from prototype research to a deployable platform. • Targets a cost-effective, maintainable design approach that lowers the barrier to adoption for research labs and future industrial pilots. • Enables the Korean SME partner to expand into the robotics sector and deliver broader societal value by translating wearable and hand-tracking expertise into a near-commercial teleoperation platform.

6. References

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(7) HaptX, https://haptx.com

(8) Manus, https://manus-meta.com

(9) SenseGlove, https://senseglove.com

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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